

Part 1: Command Line

slides adapted mainly from

<https://www.library.unlv.edu/sites/default/files/inline-images/wfOPub9DRN3jzCgLUV3ps8ksRIPVMW8alqoY7QHaPwkjPCUEXB.pdf>

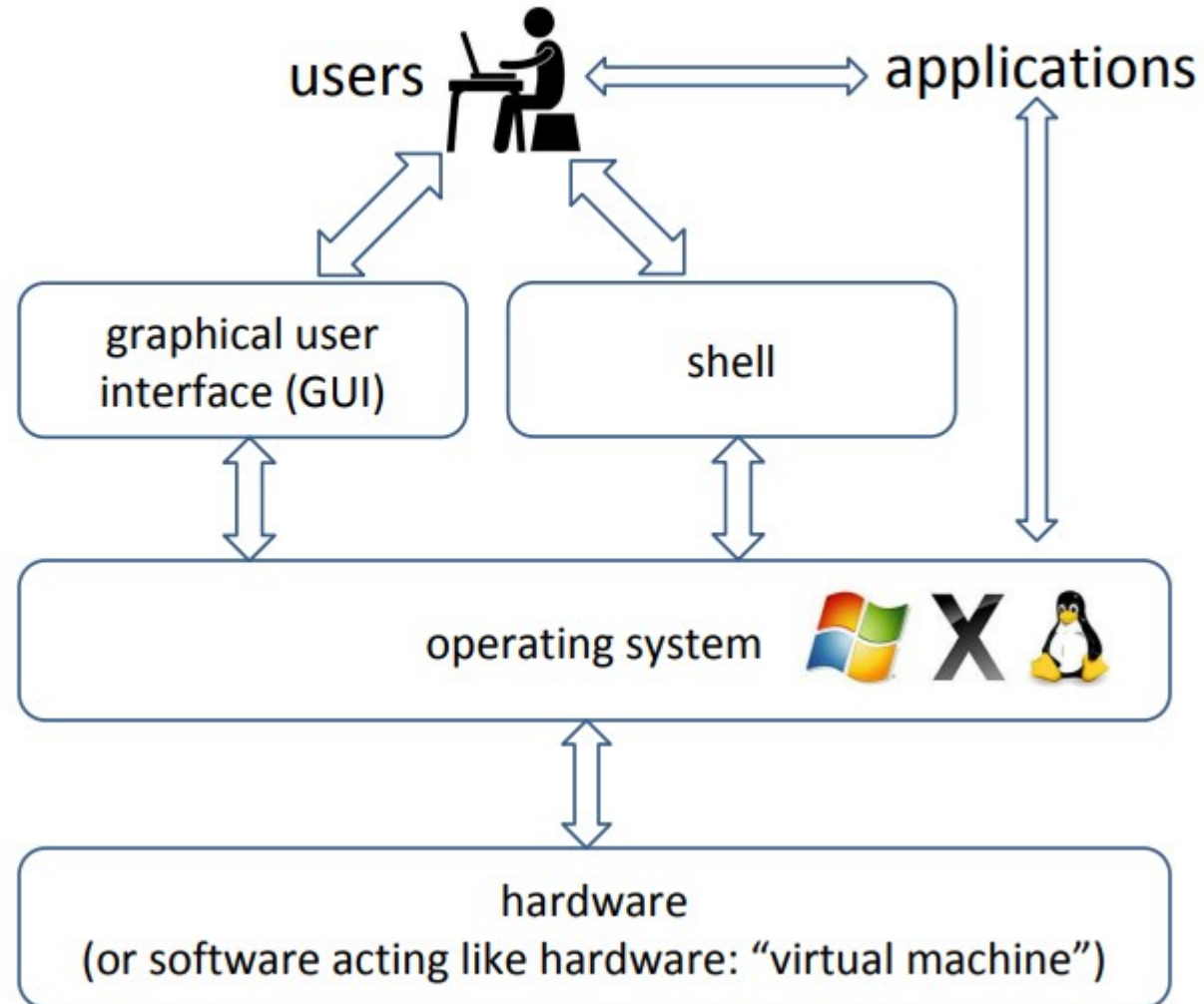
https://www2.cs.arizona.edu/classes/cs210/fall17/lectures/command_line.pdf

What is a Command Line Interface?

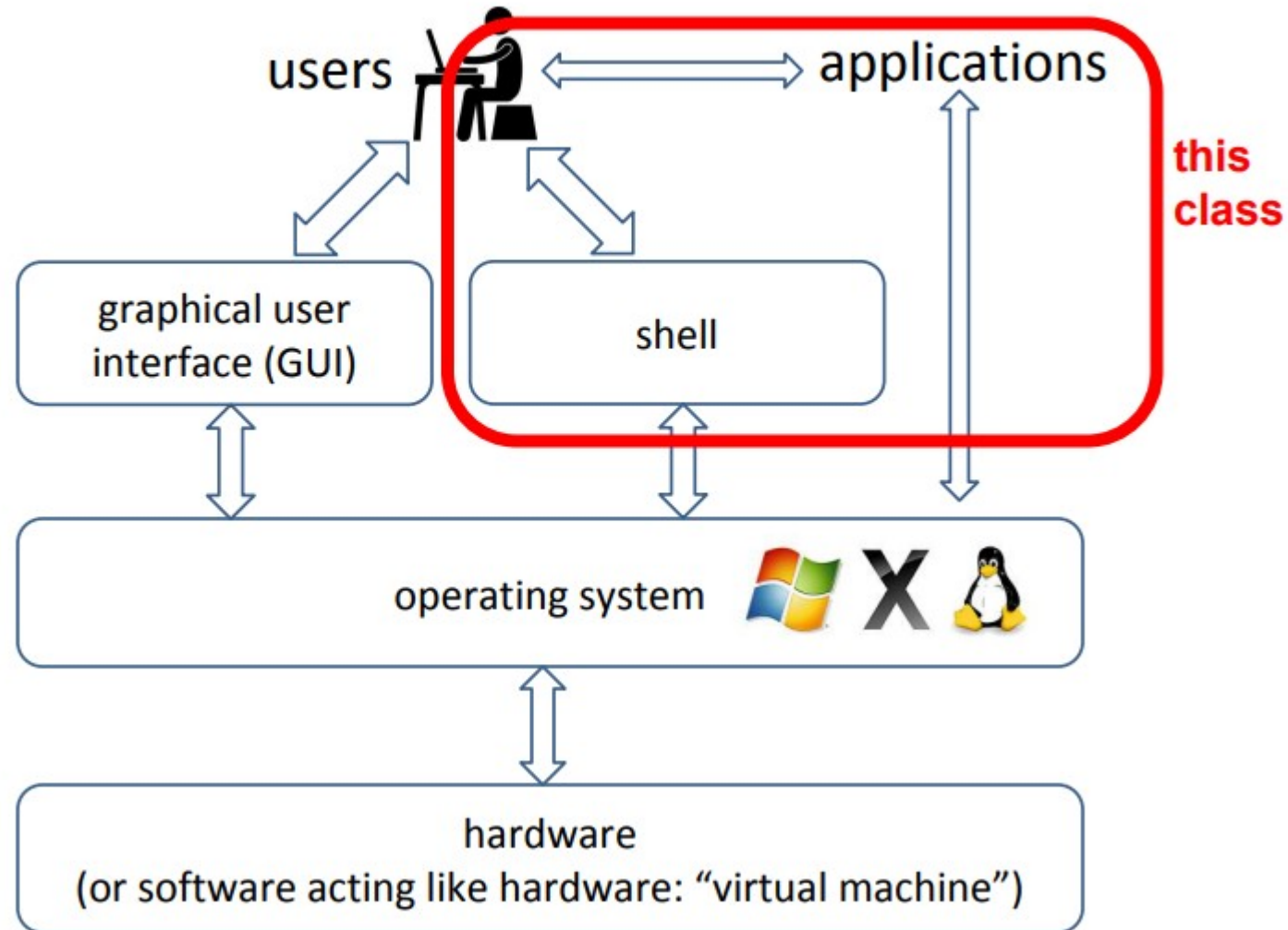
- **Interface**: Means it is a way to interact with the Operating System.
- **Command Line**: Means you interact with it through typing commands at the keyboard.

So a Command Line Interface (or a shell) is a program that lets you interact with the Operating System via the keyboard.

Organization of a computer system



Organization of a computer system



Why Use a Command Line Interface?

- A. In the old days, there was no choice
 - a. No commercial computer had a GUI until Apple released the Lisa in 1993 (at \$10, 000!!!)
 - b. There might still be no choice if you are interacting with a computer via a non-graphical terminal.
- B. Many tasks are faster than in a GUI
 - a. Suppose you wanted to see all the files in a directory that had the word “lecture” in their name.
- C. Most shells let you write scripts (programs) to automate complex tasks which you could not do with a GUI

Command Line Interface (CLI)

- **How does a command-line interface work?**
 - It begins with a **prompt** indicating the computer is ready to do something (such as `C:\>`).
 - The user types in a command and presses ENTER.
 - The command is executed.
 - A new prompt is displayed—ready for the next command.
 - CLI executes commands just as the Windows GUI does.
 - In CLI, you type the command and press ENTER.
 - In GUI, you point and click to execute commands.

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Guide to
**Managing and
Troubleshooting PCs**
Fourth Edition



```
C:\Windows\system32\cmd.exe

C:\>dir
Volume in drive C has no label.
Volume Serial Number is F4BD-E8C8

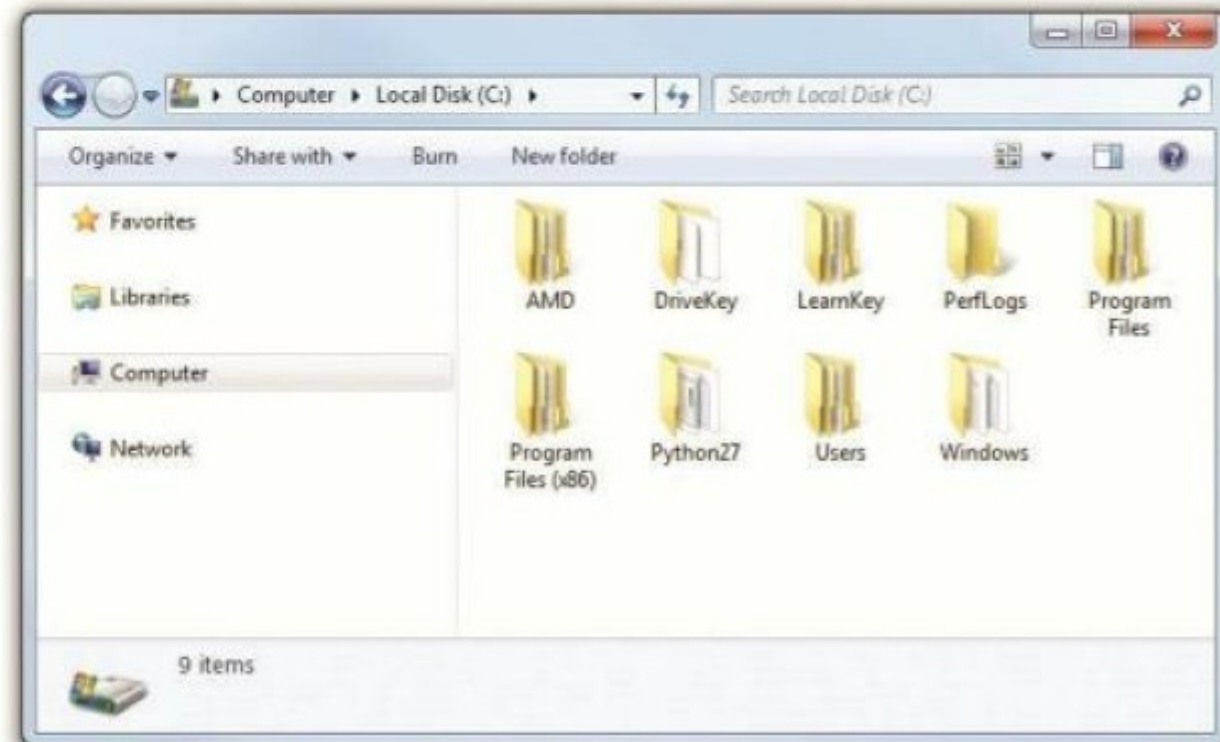
Directory of C:\

03/27/2012  09:19 AM    <DIR>          AMD
07/22/2011  09:20 AM    <DIR>          DriveKey
06/27/2011  01:32 PM    <DIR>          LearnKey
07/13/2009  10:20 PM    <DIR>          PerfLogs
04/02/2012  09:25 AM    <DIR>          Program Files
03/29/2012  08:54 AM    <DIR>          Program Files (x86)
02/27/2012  05:00 PM    <DIR>          Python27
10/05/2011  12:04 PM    <DIR>          Users
03/28/2012  08:33 AM    <DIR>          Windows
               0 File(s)              0 bytes
               9 Dir(s)  447,131,787,264 bytes free

C:\>
```

Figure 1: Contents of C: directory from the command line

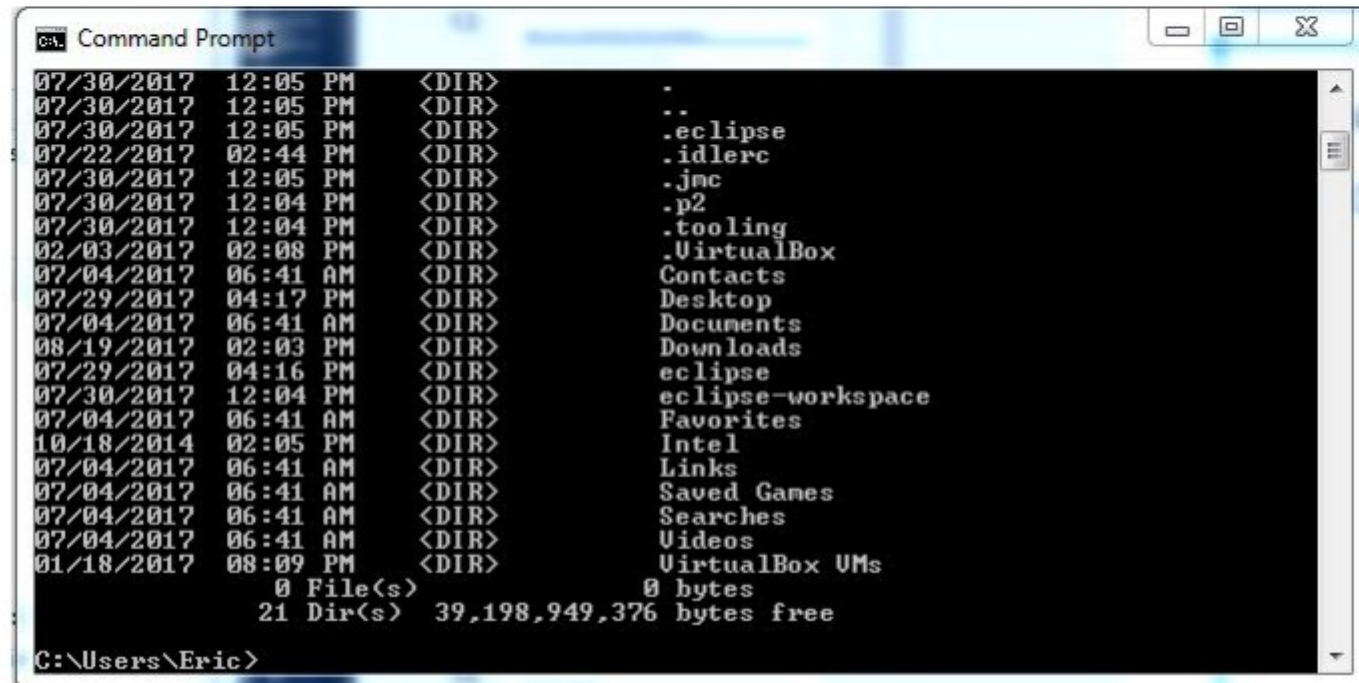
Figure 2: Contents of C: in Computer—Icon view



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Command Prompt

All versions of Windows have included a Command Prompt program. It acts like a MSDOS interface to the computer.



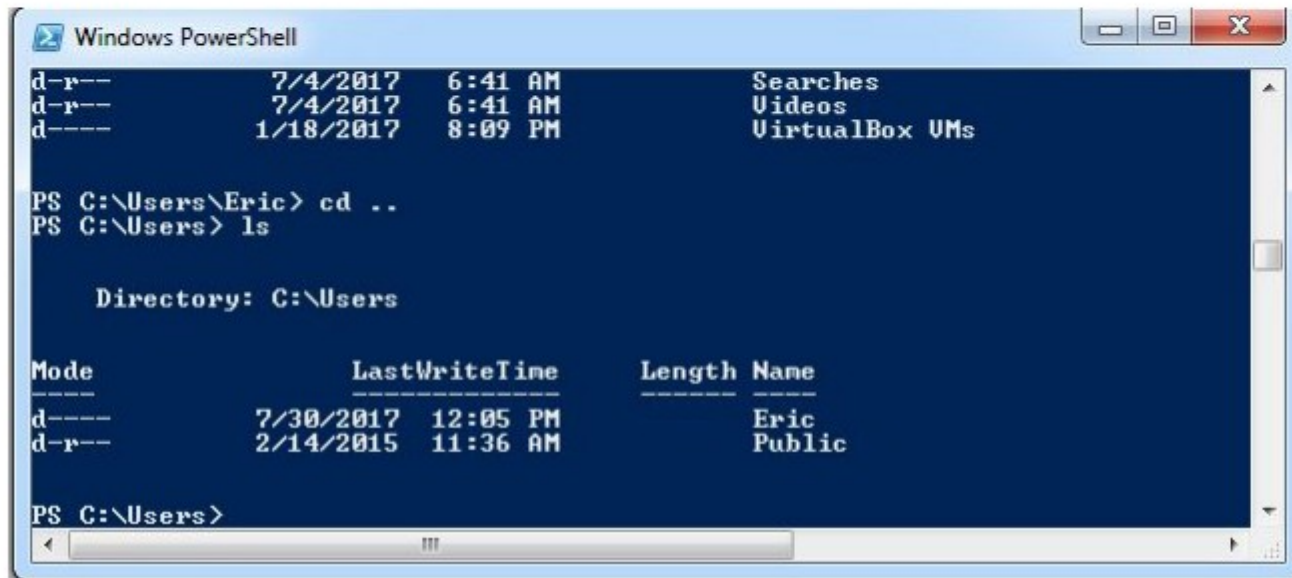
```

C:\>dir
07/30/2017 12:05 PM <DIR> .
07/30/2017 12:05 PM <DIR> ..
07/30/2017 12:05 PM <DIR> .eclipse
07/22/2017 02:44 PM <DIR> .idlerc
07/30/2017 12:05 PM <DIR> .jnc
07/30/2017 12:04 PM <DIR> .p2
07/30/2017 12:04 PM <DIR> .tooling
02/03/2017 02:08 PM <DIR> .VirtualBox
07/04/2017 06:41 AM <DIR> Contacts
07/29/2017 04:17 PM <DIR> Desktop
07/04/2017 06:41 AM <DIR> Documents
08/19/2017 02:03 PM <DIR> Downloads
07/29/2017 04:16 PM <DIR> eclipse
07/30/2017 12:04 PM <DIR> eclipse-workspace
07/04/2017 06:41 AM <DIR> Favorites
10/18/2014 02:05 PM <DIR> Intel
07/04/2017 06:41 AM <DIR> Links
07/04/2017 06:41 AM <DIR> Saved Games
07/04/2017 06:41 AM <DIR> Searches
07/04/2017 06:41 AM <DIR> Videos
01/18/2017 08:09 PM <DIR> VirtualBox VMs
               0 File(s)              0 bytes
              21 Dir(s) 39,198,949,376 bytes free

C:\Users\Eric>
```


Windows Powershell

Windows Powershell was an improved shell for Windows first released in 2006. The latest version came out in 2016.

A screenshot of a Windows PowerShell window. The window title is "Windows PowerShell". The command prompt shows a directory listing of the C:\Users directory. The output shows three entries: Searches, Videos, and VirtualBox VMs. The user then navigates to the parent directory (C:\Users) and runs the 'ls' command. The output shows a table with columns: Mode, LastWriteTime, Length, and Name. The entries are Eric and Public.

```
Windows PowerShell
d-r--      7/4/2017   6:41 AM      Searches
d-r--      7/4/2017   6:41 AM      Videos
d-----  1/18/2017   8:09 PM      VirtualBox VMs

PS C:\Users\Eric> cd ..
PS C:\Users> ls

Directory: C:\Users

Mode                LastWriteTime         Length Name
----                -
d-----      7/30/2017   12:05 PM             Eric
d-r--        2/14/2015   11:36 AM             Public

PS C:\Users>
```

- Use the **Run dialog box** or **Start Search text box**

- Start | Run
- Type **cmd**
(or)
- Type **command**
- Either runs the cmd.exe executable program found in %systemroot%\system32

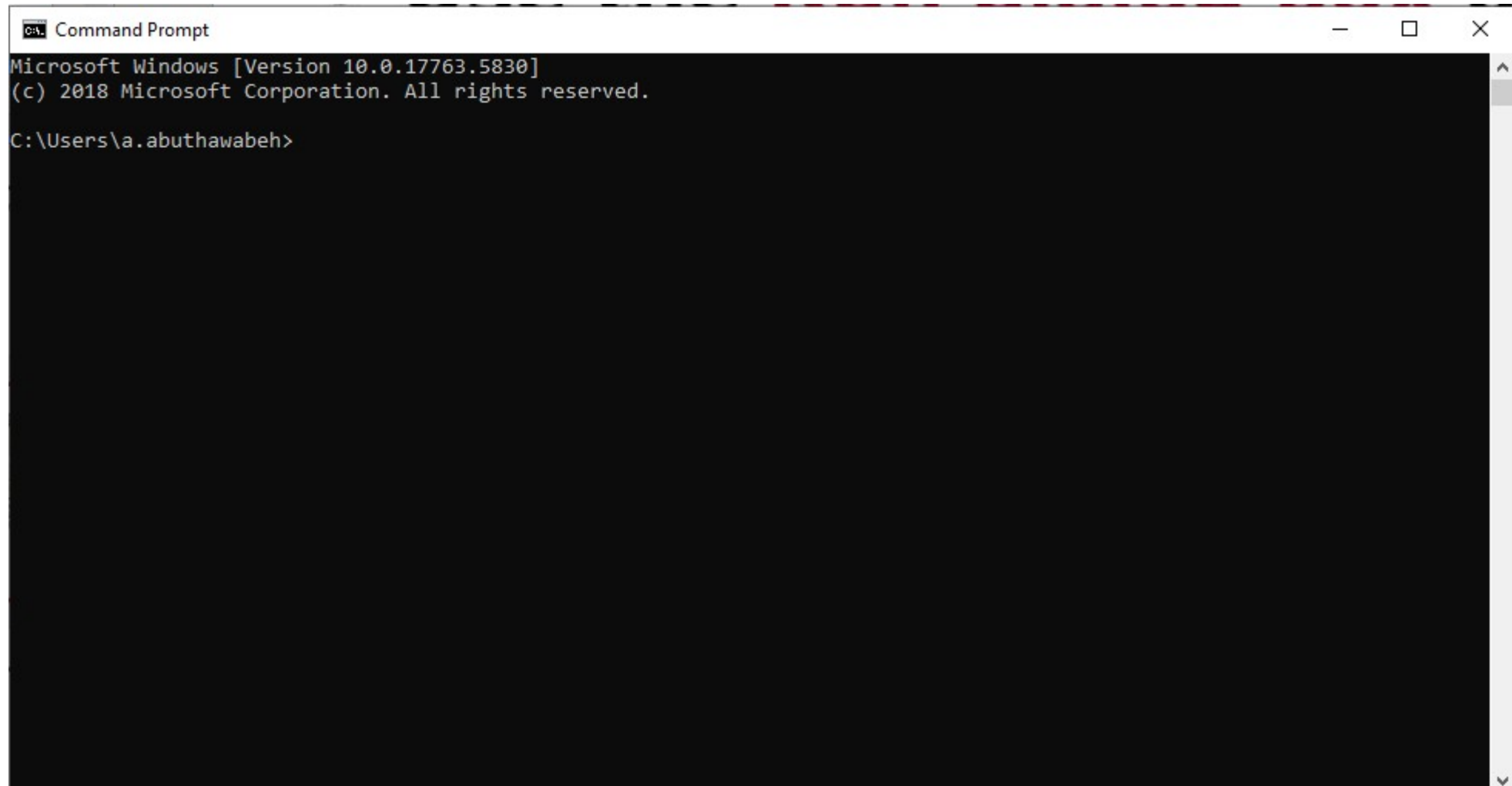


Figure 5: Type cmd in the Run dialog box to open a command-line interface window in Windows XP.

- You can also access the command line through the **Start | All Programs menu.**

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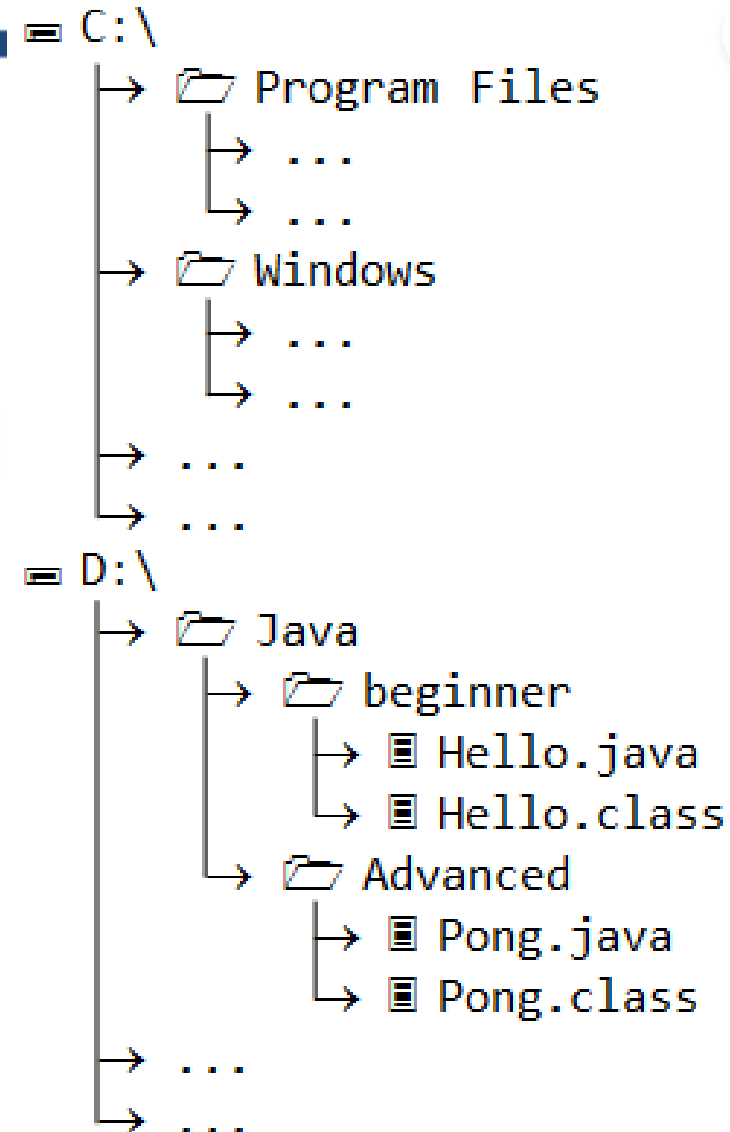


```
Command Prompt
Microsoft Windows [Version 10.0.17763.5830]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\a.abuthawabeh>
```

The file system

- Collections of files are grouped into directories (folders)
- A directory is itself a file
 - file system has a hierarchical structure (i.e., like a tree)
 - the root is referred to as "/"



- **Each program or piece of data is stored as a file on the drive.**
- **Filenames have two parts:**
 - **Filename**
 - In DOS, up to 8 characters long
 - **Extension**
 - In DOS, up to 3 characters long
 - Optional
- **The filename and extension are separated by a dot**
 - **Called the 8.3 naming system**
- **These characters cannot be used today:**
/ \ < > | : " * ?

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- **Windows does not restrict the filename to 8.3 (it can be up to 255 characters).**
 - To be backward-compatible with DOS, you need to follow the 8.3 standard.
 - Windows creates two filenames for every file to ensure backward-compatibility.
- **The extension tells the computer the type of file.**
 - .exe, .doc, .xls
 - .gif, .jpg, .png
 - .chm (help file)

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FileHomeShareViewManageLocal Disk (E:)

Drive Tools

Pin to Quick accessCopyPasteCutCopy pathPaste shortcutMove toCopy toDeleteRenameNew folderNew itemEasy accessPropertiesOpenSelect allSelect noneInvert selection

ClipboardOrganizeNewOpenSelect

←→↕↑

This PC > Local Disk (E:) >

Quick access

DesktopDownloadsDocumentsPicturesbooksmid-2023-2024-secworkshopsبارات الاتصال والكتابة

This PC

3D ObjectsDesktopDocumentsDownloadsMusicPicturesVideosLocal Disk (C:)Local Disk (D:)Local Disk (E:)

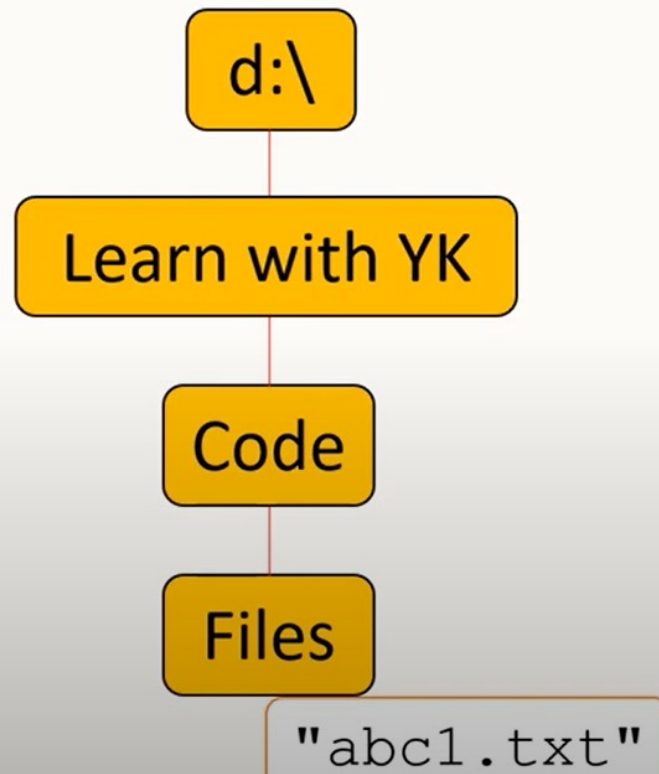
Name	Date modified	Type	Size
OneDrive_2024-04-03	4/3/2024 12:51 PM	WinRAR ZIP archive	69,645 KB
جميع الكتب - SE - طلب الكليات لشراء مصادر قسم هندسة البرمجيات معلومات	3/10/2024 3:22 PM	Microsoft Word Document	229 KB
Course_Binder_BSc_40254107_DrHossam_20222- English	9/2/2023 12:13 PM	WinRAR ZIP archive	26,356 KB
AAU Course Assessment Report Template - Update 2023 v3.0 - 120 Courses-database	6/25/2023 10:43 AM	Microsoft Excel Macro-Enabled Worksheet	619 KB
Course_Binder_BSc_sample	6/4/2023 11:05 AM	WinRAR archive	2,028 KB
CAR-V(c)_SE_40354208_Software_Development_and_Documentation	4/3/2023 9:23 AM	Microsoft Excel Macro-Enabled Worksheet	612 KB
openjdk-11+28_windows-x64_bin	4/2/2023 11:41 AM	WinRAR ZIP archive	183,005 KB
Course_Syllabus_40354208_Software_Development_and_Documentation1-AAU-2022-...	3/26/2023 11:29 AM	Microsoft Word Document	617 KB
Amman Arab uni courses	5/22/2024 11:13 AM	File folder	
workshops	5/18/2024 1:55 PM	File folder	
AAU admin documents	4/30/2024 12:07 PM	File folder	
تدقيق مساقات	4/3/2024 10:45 AM	File folder	
ملف المساق الفصل الاول 2024-2023	2/5/2024 8:19 AM	File folder	
New folder (4)	1/27/2024 1:53 PM	File folder	
New folder (3)	1/23/2024 12:00 PM	File folder	
syl	11/19/2023 2:05 PM	File folder	
OracleXE213_Win64 (1)	11/15/2023 2:50 PM	File folder	
exper	11/14/2023 8:14 AM	File folder	
New folder (2)	10/25/2023 10:59 ...	File folder	
reserach - AAU	9/26/2023 9:00 AM	File folder	
test-exper1	9/10/2023 8:41 AM	File folder	
ملف المساق الفصل الثاني -2022-2023	7/8/2023 8:17 AM	File folder	
New folder	6/26/2023 2:40 PM	File folder	

1. An absolute path refers to the same fixed location, whereas a relative path refers to a location which is relative to the current working directory.
2. An absolute path always starts with the root directory in a drive (C:\, D:\, etc.), whereas a relative path never starts with drive name.

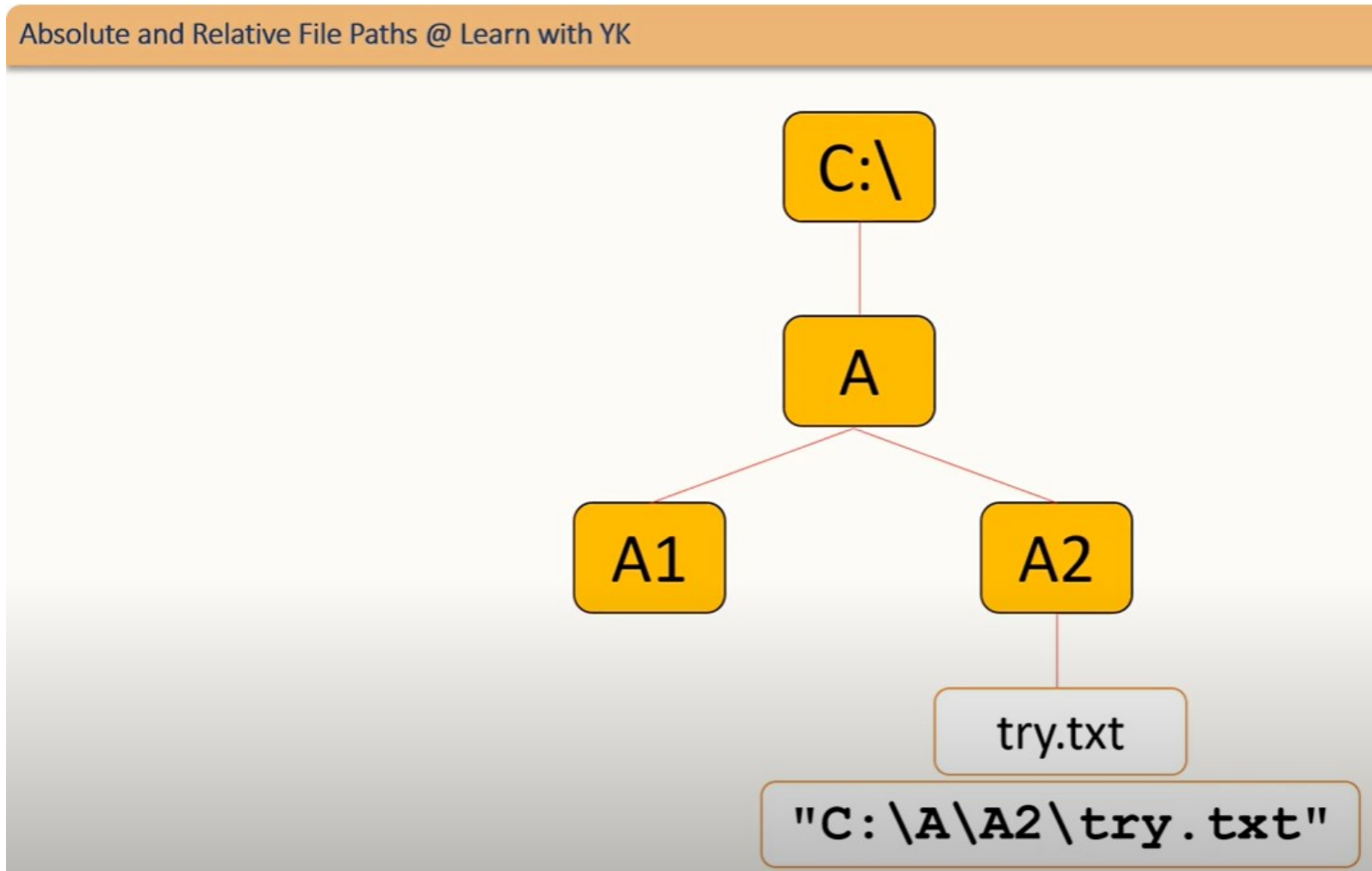
Absolute path

"abc1.txt"

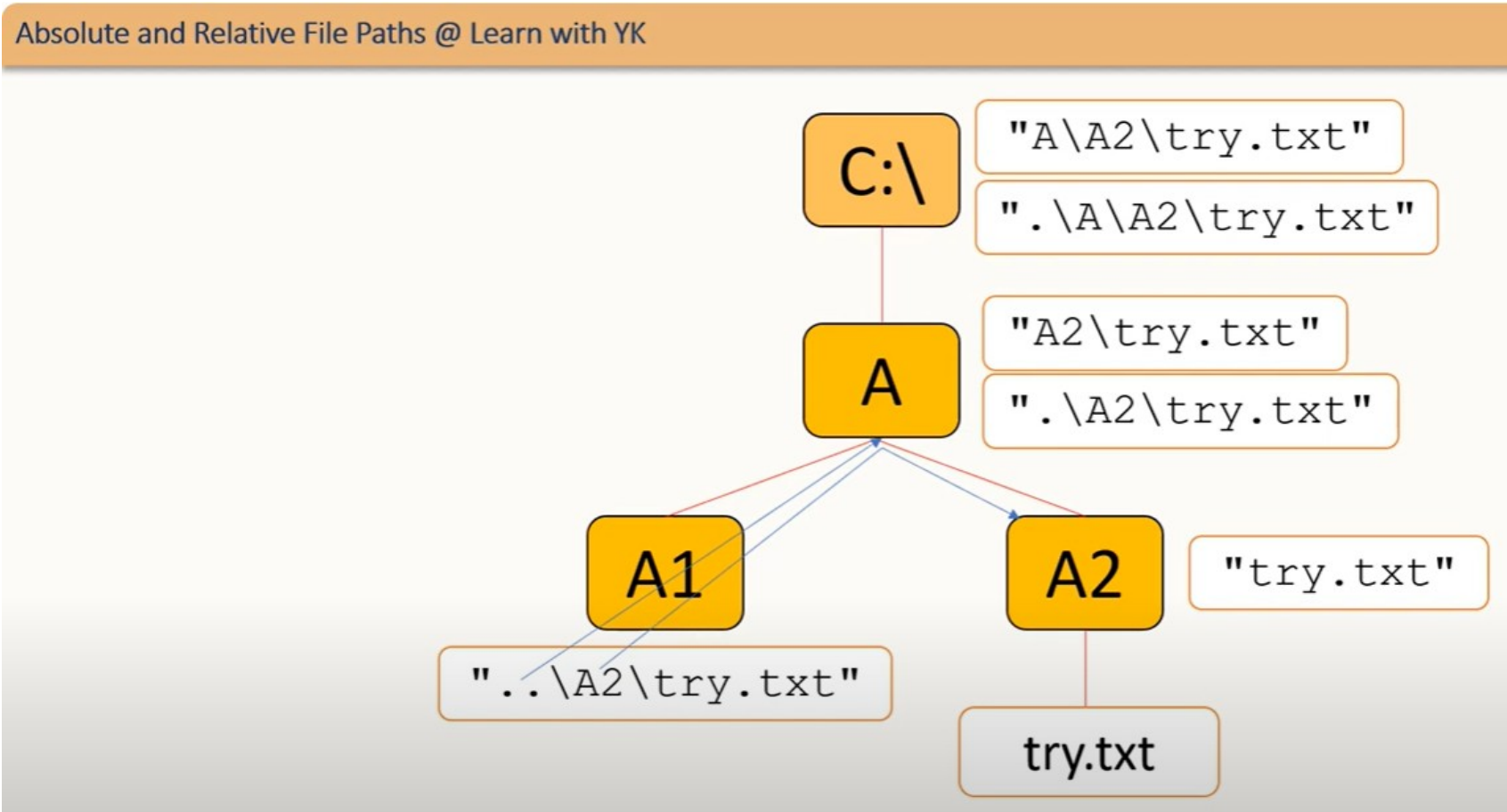
"d:\Learn with YK\Code\Files\abc1.txt"



Absolute path

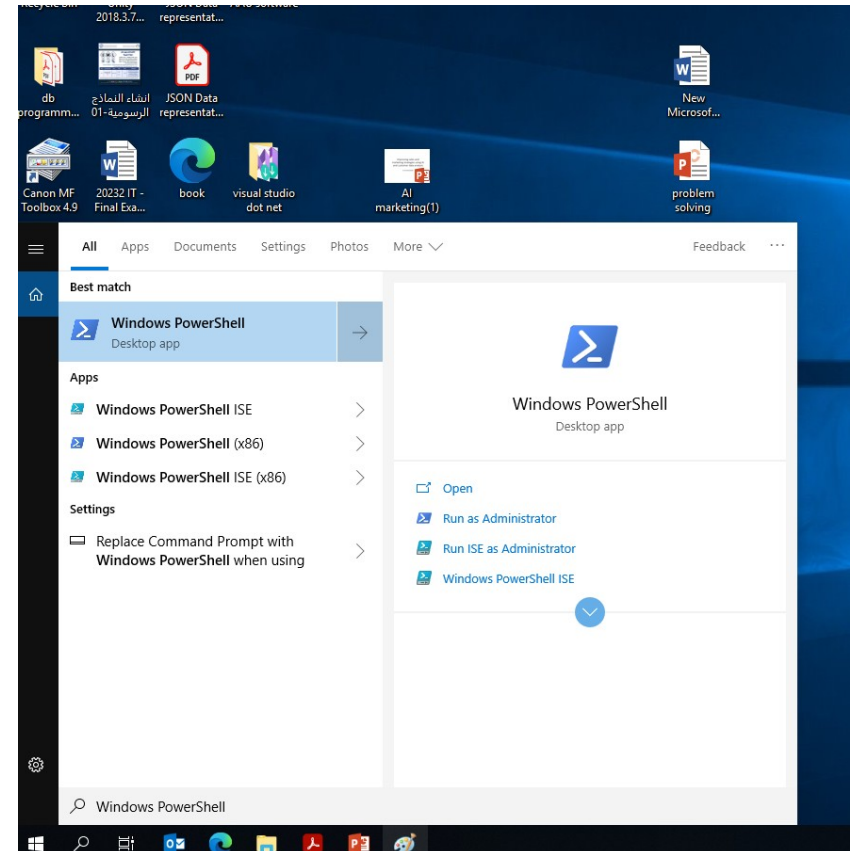


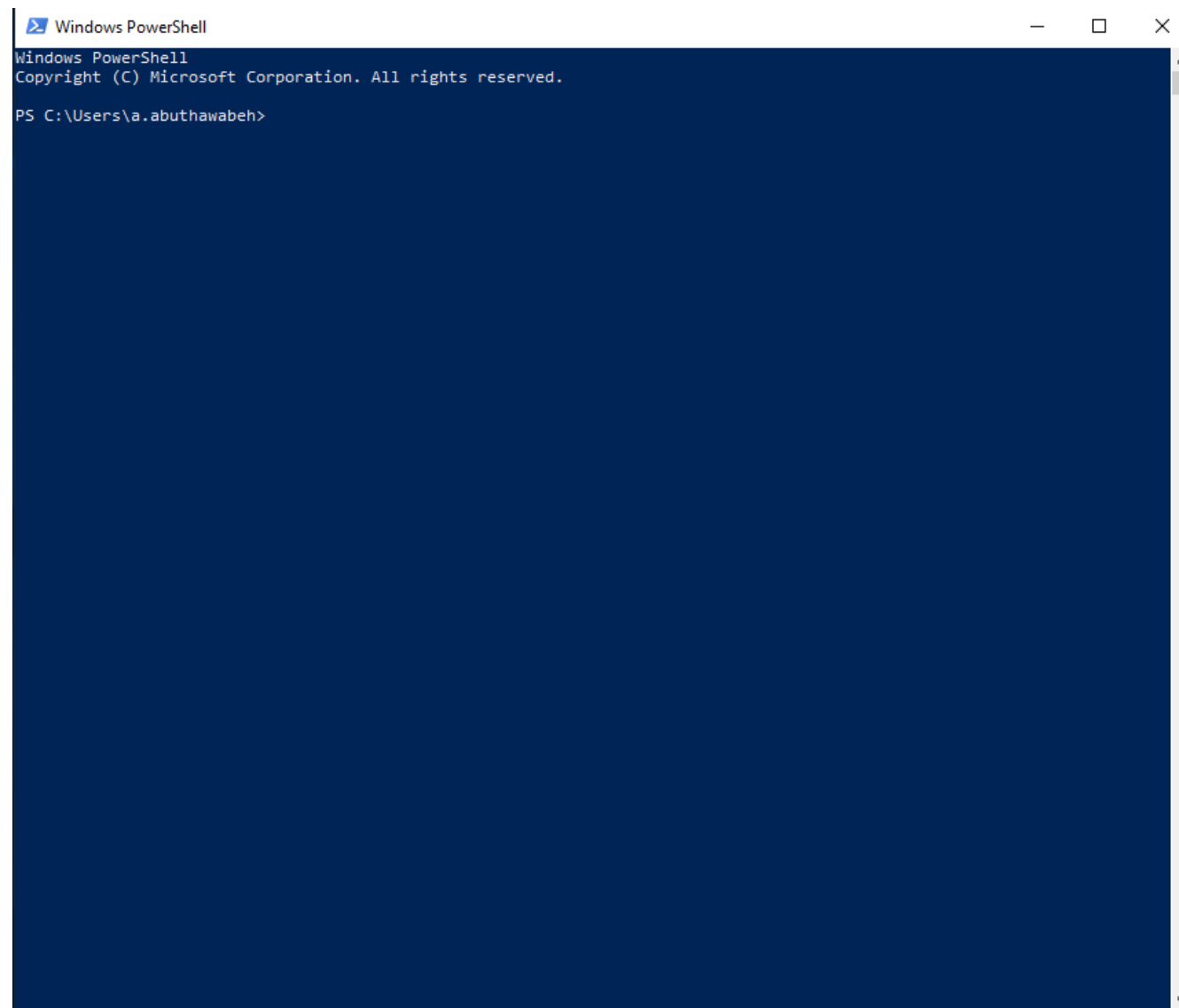
Relative path



Let's get started by opening PowerShell

- To open PowerShell, use the taskbar to search for PowerShell and select 'Windows PowerShell'



A screenshot of a Windows PowerShell console window. The window has a dark blue background and a white title bar at the top. The title bar contains the text "Windows PowerShell" and standard window control buttons (minimize, maximize, close). The main area of the window is dark blue and contains white text. The text reads: "Windows PowerShell", "Copyright (C) Microsoft Corporation. All rights reserved.", and "PS C:\Users\A.abuthawabeh>". There is a vertical scrollbar on the right side of the window.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\A.abuthawabeh>
```

Review

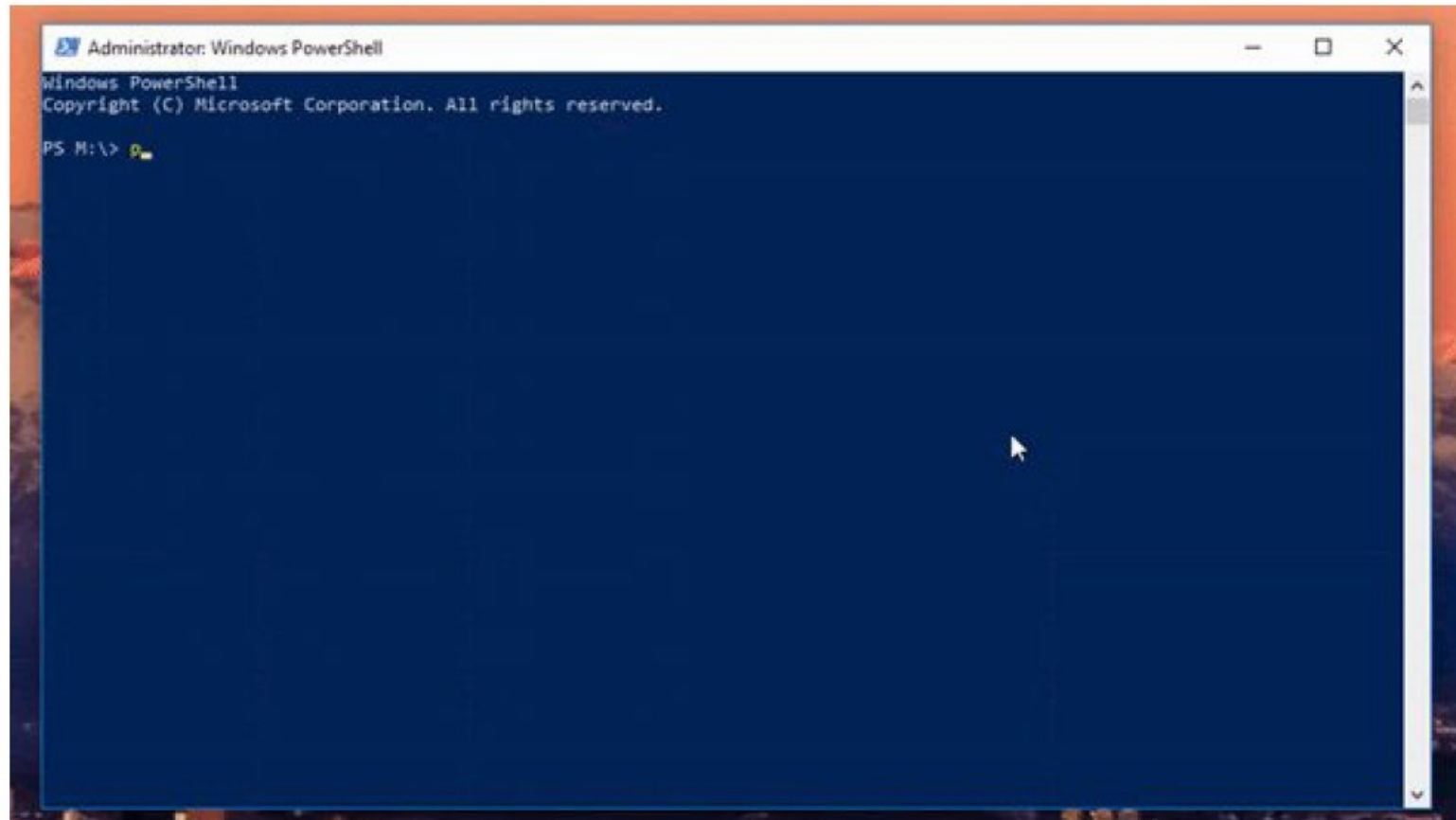
Reference the table below to review the material we covered in this tutorial.

Command:	Stands for:	Does what?
<code>pwd</code>	Print working directory	Lets you know where you are in your file system
<code>ls</code>	List files	Lists the files in your current directory
<code>cd</code>	Change directory	Changes the current directory
<code>cd ..</code>	Navigate up one directory	Navigate up one directory from the current directory
<code>cd ../..</code>	Navigate up two directories	Navigate up two directories from the current directory
<code>mkdir</code>	Make directory	Creates a directory
<code>mv</code>	Move	Moves a file, can also be used to rename a file
<code>rm</code>	Remove	Deletes a file
<code>cat</code>	Concatenate	Will read a file
<code>exit</code>	Exit	Exit PowerShell

Moving Around Using PowerShell

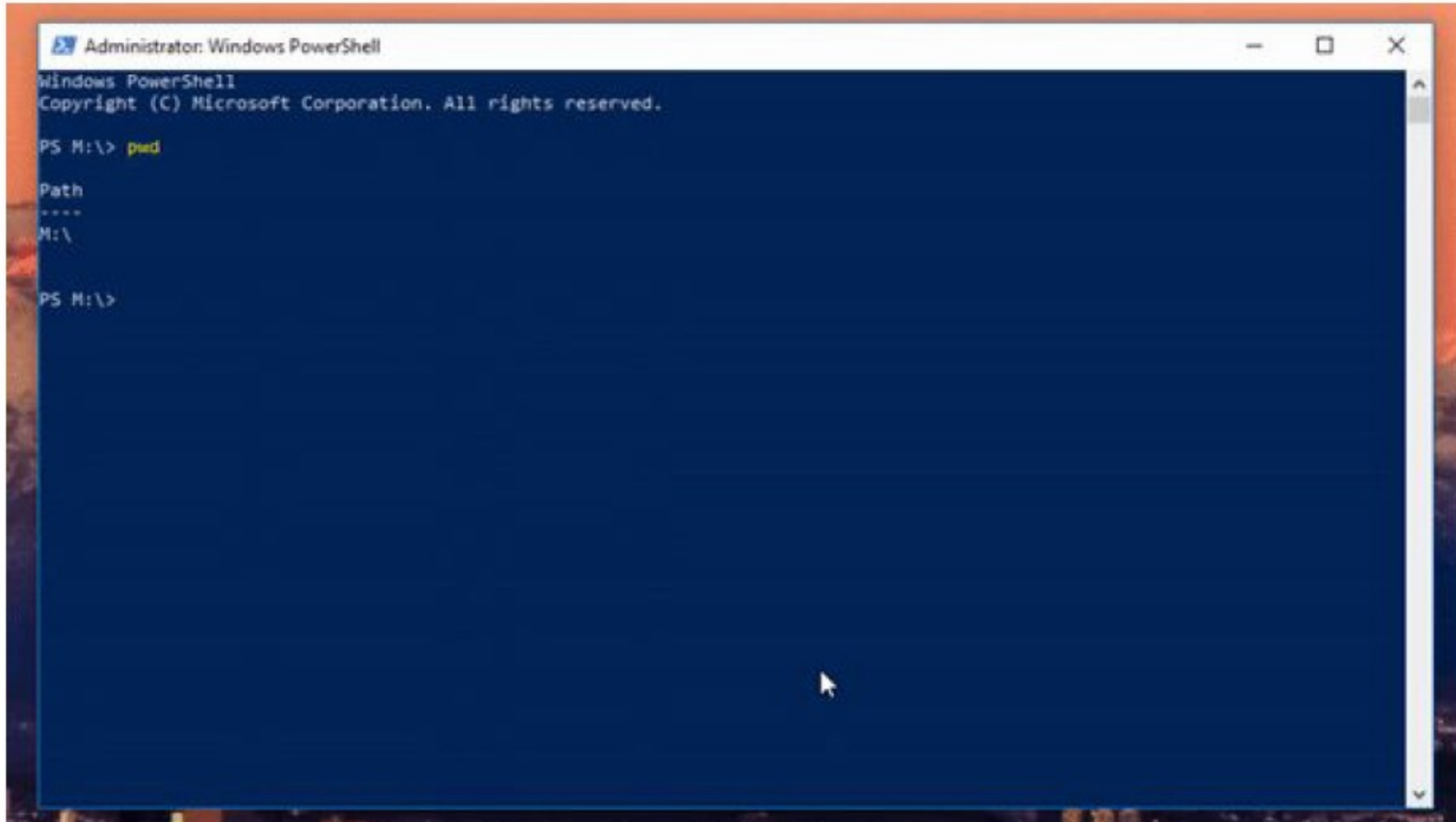
If you are feeling lost after opening PowerShell, it is important to find out where you are in your filesystem. Go ahead and enter the following command to **print the working (your current) directory**:

```
pwd
```



Nice job! Now let's see which directories and files are available in your location by entering the **list files** command:

ls

A screenshot of a Windows PowerShell terminal window titled "Administrator: Windows PowerShell". The window has a dark blue background. The text displayed is as follows:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS M:\> pwd

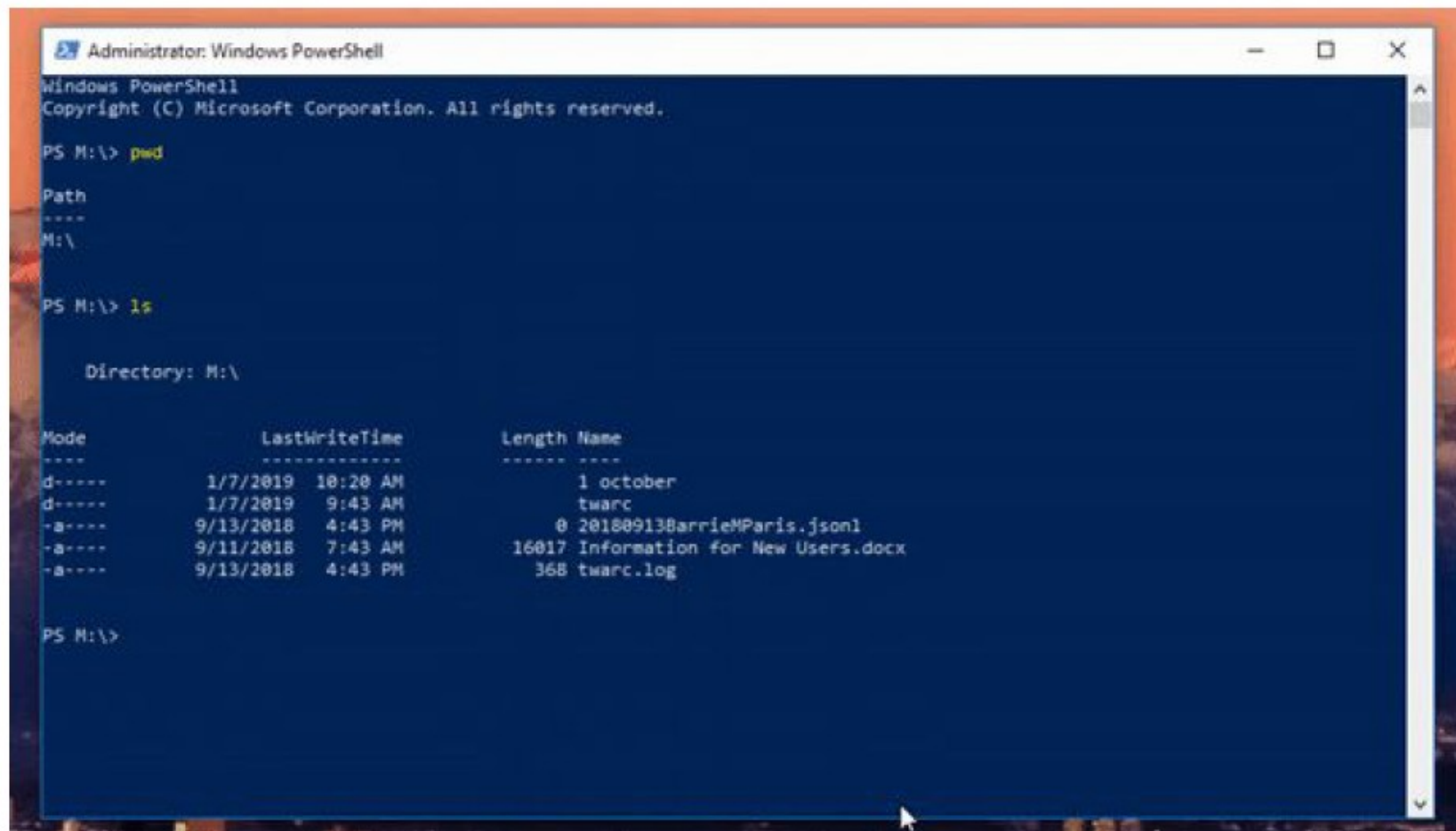
Path
----
M:\

PS M:\>
```

A mouse cursor is visible near the bottom center of the terminal window.

Let's change directories. Navigate to your Desktop by entering the **change directory** command:

```
cd D:\Desktop
```



The screenshot shows an Administrator Windows PowerShell window with a dark blue background. The window title is "Administrator: Windows PowerShell". The prompt is "PS M:\>". The user has entered the command "pwd", and the output is "Path M:\\". The user has then entered the command "ls", and the output is a directory listing for "Directory: M:\\". The listing shows a table with columns for Mode, LastWriteTime, Length, and Name. The files listed are "1 october", "twarc", "20180913BarrieMParis.json1", "Information for New Users.docx", and "twarc.log".

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS M:\> pwd

Path
----
M:\

PS M:\> ls

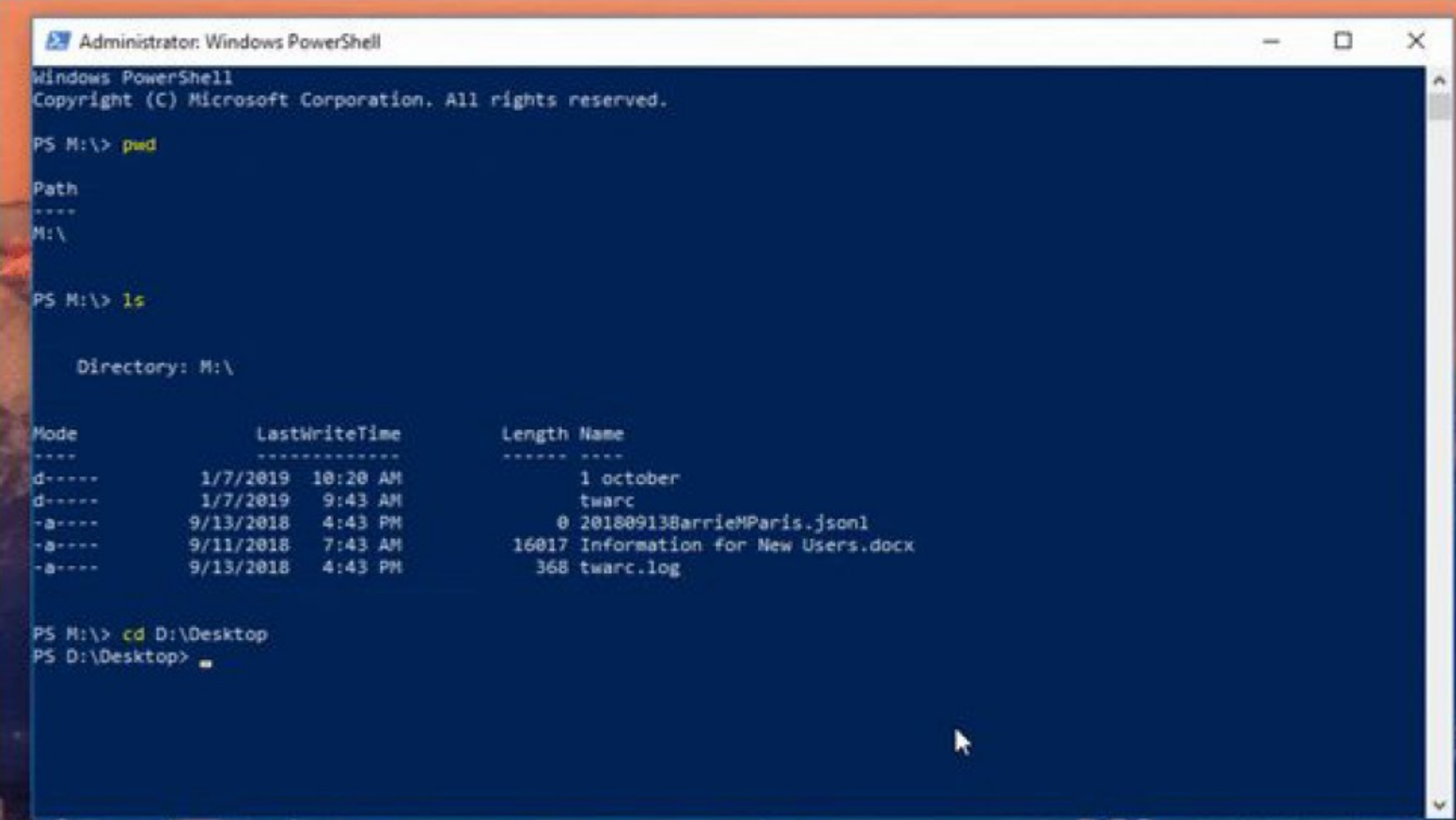
Directory: M:\

Mode                LastWriteTime         Length Name
----                -
d-----          1/7/2019 10:20 AM             1 1 october
d-----          1/7/2019  9:43 AM             1 twarc
-a-----          9/13/2018  4:43 PM             0 20180913BarrieMParis.json1
-a-----          9/11/2018  7:43 AM        16017 Information for New Users.docx
-a-----          9/13/2018  4:43 PM          368 twarc.log

PS M:\>
```


let's create a directory by entering the **make directory** command:

```
mkdir walt_whitman
```



The screenshot shows a Windows PowerShell window titled "Administrator: Windows PowerShell". The window has a dark blue background with white text. The user has entered the following commands:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS M:\> pwd

Path
----
M:\

PS M:\> ls

Directory: M:\

Mode                LastWriteTime         Length Name
----                -
d-----          1/7/2019 10:20 AM             1 october
d-----          1/7/2019  9:43 AM             twarc
-a-----          9/13/2018  4:43 PM             0 20180913BarrieMParis.jsonl
-a-----          9/11/2018  7:43 AM          16017 Information for New Users.docx
-a-----          9/13/2018  4:43 PM             368 twarc.log

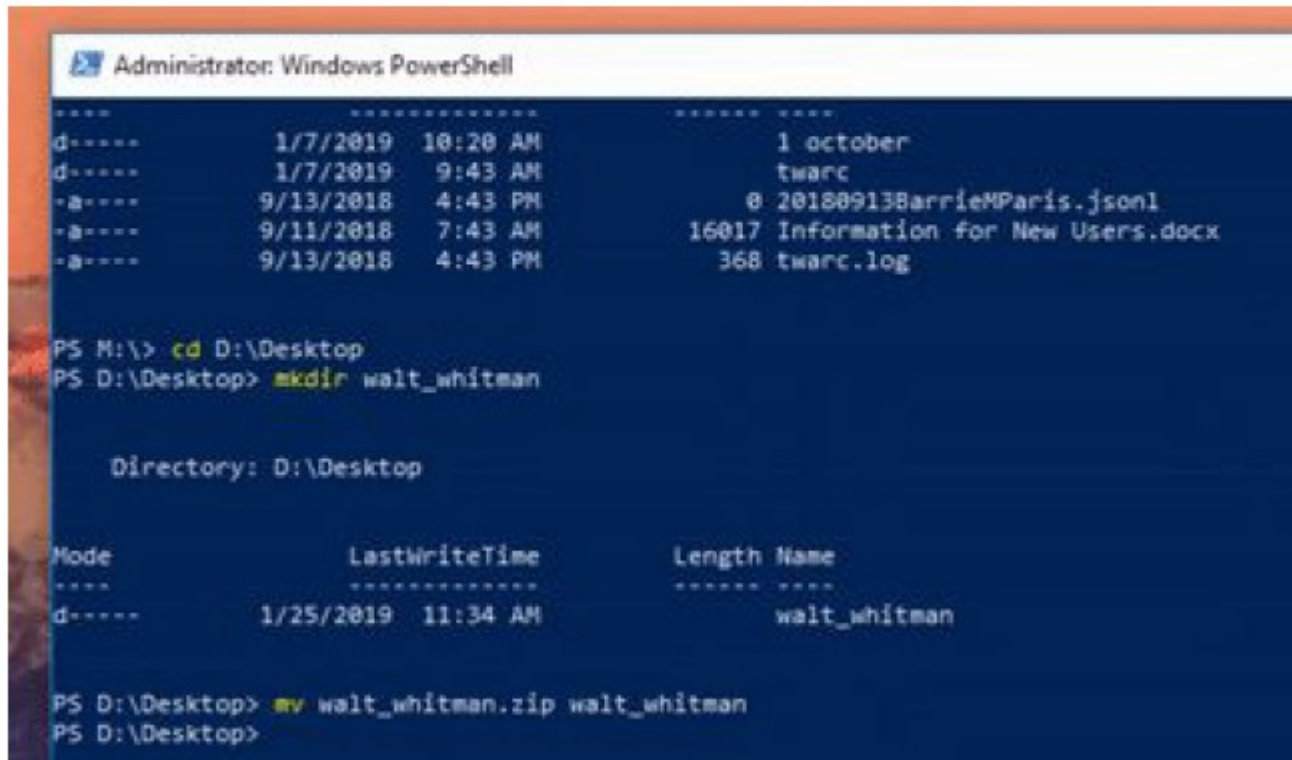
PS M:\> cd D:\Desktop
PS D:\Desktop>
```

The output of the `ls` command shows a directory listing for `M:\`. The listing includes columns for Mode, LastWriteTime, Length, and Name. The files listed are `october`, `twarc`, `20180913BarrieMParis.jsonl`, `Information for New Users.docx`, and `twarc.log`.

We're going to move the 'walt_whitman' zip file you downloaded from your Desktop into your 'walt_whitman' directory by using a new command, **move**:

```
mv walt_whitman.zip walt_whitman
```

```
cd D:\Desktop\walt_whitman
```



The screenshot shows an Administrator Windows PowerShell window with a dark blue background. The title bar reads 'Administrator: Windows PowerShell'. The window displays the following commands and their outputs:

```
PS M:\> cd D:\Desktop
PS D:\Desktop> mkdir walt_whitman

Directory: D:\Desktop

Mode                LastWriteTime         Length Name
----                -
d-----           1/25/2019  11:34 AM                walt_whitman

PS D:\Desktop> mv walt_whitman.zip walt_whitman
PS D:\Desktop>
```

At the top of the window, there is a directory listing for 'D:\Desktop' showing files like '1 october', 'twarc', '0 20180913BarrieMParis.jsonl', '16017 Information for New Users.docx', and '368 twarc.log'.

Let's check out the 'readme.txt' file to see if it tells us about any of the other poems. You can open the file by using the **cat** (concatenate) command:

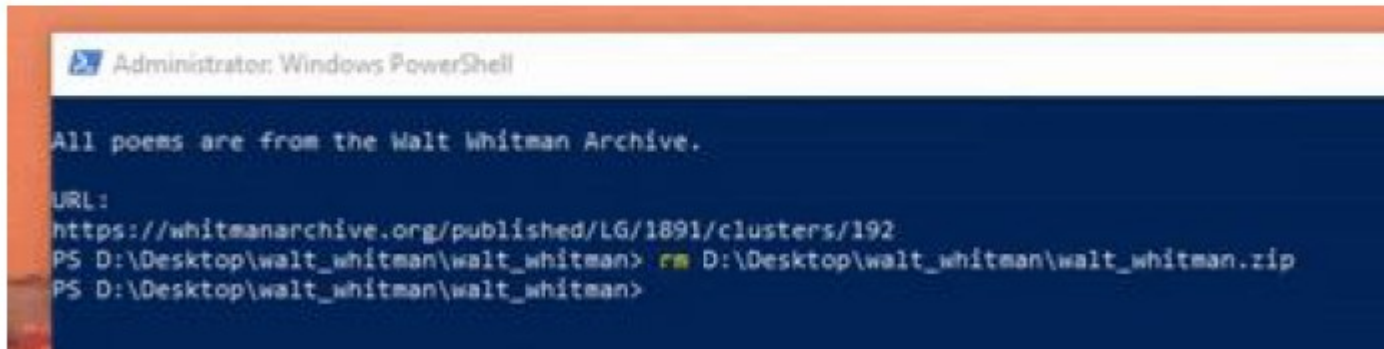
```
cat readme.txt
```

Before we exit PowerShell, let's get rid of the 'walt_whitman.zip' file since we don't need it anymore. Do this by entering the **remove** command:

```
rm D:\Desktop\walt_whitman\walt_whitman.zip
```

Since we're all done in this directory, navigate up two directories using the **change directory** command:

```
cd ../../
```

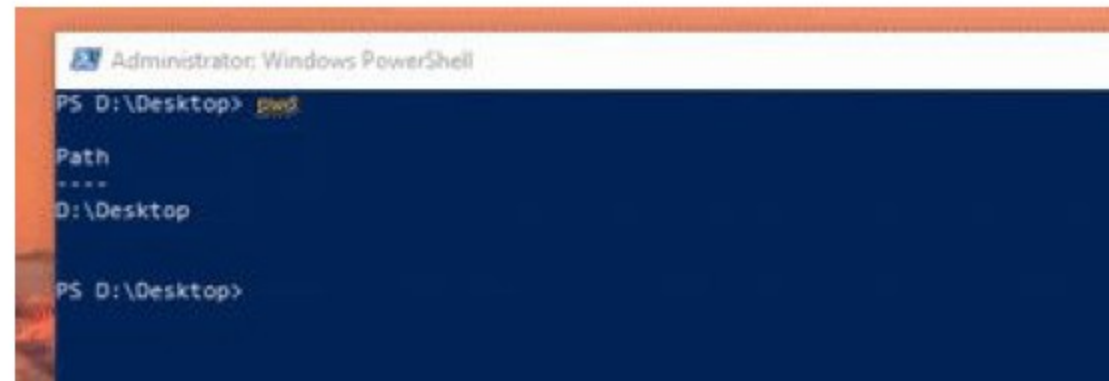


```
Administrator: Windows PowerShell

All poems are from the Walt Whitman Archive.

URL:
https://whitmanarchive.org/published/LG/1891/clusters/192
PS D:\Desktop\walt_whitman\walt_whitman> rm D:\Desktop\walt_whitman\walt_whitman.zip
PS D:\Desktop\walt_whitman\walt_whitman>
```

```
exit
```



```
Administrator: Windows PowerShell

PS D:\Desktop> exit

Path
----
D:\Desktop

PS D:\Desktop>
```



Windows Command	Linux Command	Explanation
dir	ls	List files in current directory
copy	Cp	Copy a file
move	mv	Move a file
ren	mv	Rename a file
del	rm	Delete a file
mkdir	mkdir	Create a new folder
<u>cls</u>	clear	Clear the terminal screen
find	Grep	Search strings in a file
exit	exit	Close the terminal
cd	cd	Change directory location

Compare linux with windows commands examples
<https://angrytools.com/command/>

Linux commands categories



Brij Kishore Pandey

Don't Forget to
Save For Later

LINUX COMMAND CHEATSHEET

Files & Navigation

ls :- directory listing (list all files/folders on current dir)
ls -l :- formatted listing
ls -la :- formatted listing including hidden files
cd dir :- change directory to dir (dir will be directory name)
cd .. :- change to parent directory
cd ../dir :- change to dir in parent directory
cd :- change to home directory
pwd :- show current directory
mkdir dir - create a directory dir
rm file :- delete file
rm -f dir :- force remove file
rm -r dir :- delete directory dir
rm -rf dir :- remove directory dir
cp file1 file2 :- copy file1 to file2
mv file1 file2 :- rename file1 to file2
mv file1 dir/file2 :- move file1 to dir as file2
touch file :- create or update file
cat file :- output contents of file
cat > file :- write standard input into file
cat » file :- append standard input into file
tail -f file :- output contents of file as it grows

Networking

ping host :- ping host
whois domain :- get whois for domain
dig domain :- get DNS for domain
dig -x host :- reverse lookup host
wget file :- download file
wget -c file :- continue stopped download
wget -r url :- recursively download files from url
curl url :- outputs the webpage from url
curl -o meh.html url :- writes the page to meh.html
ssh user@host :- connect to host as user
ssh -p port user@host :- connect using port
ssh -D user@host :- connect & use bind port

Processes

ps :- display currently active processes
ps aux :- detailed outputs
kill pid :- kill process with process id (pid)
killall proc :- kill all processes named proc

System Info

date :- show current date/time
uptime :- show uptime
whoami :- who you're logged in as
w :- display who is online
cat /proc/cpuinfo :- display cpu info
cat /proc/meminfo :- memory info
free :- show memory and swap usage
du :- show directory space usage
du -sh :- displays readable sizes in GB
df :- show disk usage
uname -a :- show kernel config

Archive & compress

tar cf file.tar files :- tar files into file tar
tar xf file.tar :- untar into current directory
tar tf file.tar :- show contents of archive
Options :-
c - create archive **j** - bzip2 compression
t - table of contents **w** - ask for confirmation
x - extract **k** - do not overwrite
z - use zip/gzip **T** - files from file
f - specify filename **v** - verbose

Permissions

chmod octal file - change permissions of file
4 - read (r)
2 - write (w)
1 - execute (x)
order: owner/group/world
chmod 777 :- rwx for everyone
chmod 755 :- rw for owner, rx for group world

Other Commands

grep pattern files :- search in files for pattern
grep -r pattern dir :- search for pattern recursively in dir
locate file :- find all instances of file
whereis app :- show possible locations of app
man command :- show manual page for command

Linux commands examples

NAVIGATE FILES

LIST DIRECTORIES (WITH TYPE INDICATOR)

```
$ ls --file-type
```

CHANGE DIRECTORY TO "EXAMPLE"

```
$ cd example
```

MOVE UP ONE DIRECTORY

```
$ cd ..
```

MOVE UP TWO DIRECTORIES

```
$ cd ../..
```

CHANGE TO HOME DIRECTORY

```
$ cd ~
```

GET CURRENT DIRECTORY

```
$ pwd
```

GET ABSOLUTE PATH TO A FILE OR FOLDER

```
$ readlink -f example
```

GET FILE TYPE OF "EXAMPLE.EXT"

```
$ file example.ext
```

INSTALLING SOFTWARE

- On Fedora and CentOS, [COMMAND] is dnf
- On Ubuntu and Debian, [COMMAND] is apt
- On OpenSUSE, [COMMAND] is zypper
- Other distributions may use different commands

SEARCH FOR AN APPLICATION CALLED EXAMPLE

```
$ sudo [COMMAND] search example
```

INSTALL AN APPLICATION CALLED EXAMPLE

```
$ sudo [COMMAND] install example
```

UNINSTALL AN APPLICATION CALLED EXAMPLE

```
$ sudo [COMMAND] remove example
```

SERVICES

START SERVICES

```
$ sudo systemctl start example
```

STOP SERVICES

```
$ sudo systemctl stop example
```

GET STATUS OF SERVICES

```
$ sudo systemctl status example
```

Linux commands examples

FILE MANAGEMENT

COPY A FILE IN PLACE

```
$ cp example.txt example-1.txt
```

COPY A FILE TO DOCUMENTS

```
$ cp example.txt ~/Documents/example-1.txt
```

MOVE A FILE TO DOCUMENTS

```
$ mv example.txt ~/Documents
```

CREATE A DIRECTORY (FOLDER)

```
$ mkdir example
```

REMOVE AN EMPTY DIRECTORY

```
$ rmdir example
```

SAFELY REMOVE A FILE

```
$ trash example.txt
```

REMOVE A FILE (WITHOUT TRASH COMMAND)

```
$ mv example.txt ~/.local/share/Trash/files
```

PERMANENTLY DELETE A FILE

```
$ shred example.txt
```

DOWNLOAD A FILE FROM AN NETWORK LOCATION

```
$ wget http://example.com/file
```

More Linux commands examples

File and directory commands	
Command	Action
ls	displays files/directories in 3 column format
ls -la	displays files/directories in long format, including hidden files
ls -R	displays files/directories recursively
ls -F	displays files/directories and appends indicator for file type or directory
ls --color=auto	control whether color is used to distinguish file types. (never, always or auto)
cd <i>dir</i>	change directory to <i>dir</i>
pwd	show current directory
mkdir <i>dir</i>	create a directory <i>dir</i>
rmdir <i>dir</i>	delete directory <i>dir</i>
rm <i>file</i>	delete <i>file</i>
rm -r <i>dir</i>	delete directory <i>dir</i>
rm -f <i>file</i>	force remove <i>file</i>
rm -rf <i>dir</i>	force remove <i>dir</i> (DANGEROUS)
cp <i>file1 file2</i>	copy <i>file1</i> to <i>file2</i>
cp <i>file1 file2</i>	copy <i>dir1</i> to <i>dir2</i> ; create <i>dir2</i> if it doesn't exist
mv <i>file1 file2</i>	rename or move <i>file1</i> to <i>file2</i> if <i>file2</i> is an existing directory, moves <i>file1</i> into directory <i>file2</i>
touch <i>file</i>	create or update <i>file</i>
cat <i>file</i>	displays contents of <i>file</i>
less <i>file</i>	displays contents of <i>file</i> , allows for forward/reverse navigation of file
head <i>file</i>	output the first 10 lines of <i>file</i>
tail <i>file</i>	output the last 10 lines of <i>file</i>
tail -f <i>file</i>	output the last 10 lines of <i>file</i> and output continues as <i>file</i> is being updated, to end CTRL+C
grep <i>word</i> or "<i>phrase</i>" <i>file</i>	search <i>file</i> for <i>word</i> or " <i>phrase</i> " NOTE: case sensitive. <i>Phrase</i> must be enclosed in quotes
grep <i>word</i> or "<i>phrase</i>"*	search <i>all</i> files in current directory for <i>word</i> or " <i>phrase</i> " NOTE: case sensitive. <i>Phrase</i> must be enclosed in quotes
grep -i <i>word</i> or "<i>phrase</i>" <i>file</i>	search <i>file</i> for <i>word</i> or " <i>phrase</i> " Case insensitive. <i>Phrase</i> must be enclosed in quotes
grep -i <i>word</i> or "<i>phrase</i>"*	search <i>all</i> files in current directory for <i>word</i> or " <i>phrase</i> " NOTE: case insensitive. <i>Phrase</i> must be enclosed in quotes
find . -name <i>file</i>	Starting at current directory look for the file named <i>file</i>
find <i>dir</i> -name <i>file</i>	Starting at <i>dir</i> look for the file named <i>file</i>

File Permissions

`chmod octal file` – change the permissions of *file* to *octal*, which can be found separately for user, group, and world by adding:

- 4 – read (r)
- 2 – write (w)
- 1 – execute (x)

Examples:

`chmod 777` – read, write, execute for all

`chmod 755` – rwx for owner, rx for group and world

Help commands

Command	Action
<code><i>progname --help</i> or <i>progname -h</i></code>	displays help/usage for program – if provided
<code>man <i>progname</i></code>	displays documentation for <i>progname</i>
<code>man -k <i>progname</i></code>	searches all documentation containing <i>progname</i>

System commands

Command	Action
<code>date</code>	shows the current date and time
<code>cal</code>	show this month's calendar
<code>whoami</code>	who you are logged in as
<code>df</code>	show quota information/disk usage
<code>which <i>progname</i></code>	shows path information for <i>progname</i>

Short cuts	
Command	Action
Ctrl+C	halts current command
Ctrl+D	logout, similar to exit
Ctrl+U	erases line of input at command line
!!	repeats last command
history	displays history of commands with number of command
!#	repeats # of command from history, Example: !76 will repeat the 76 th command in history
up arrow	displays last command, continually pressing up arrow will allow to scroll through previous commands
tab	auto complete

Remote Access	
Command	Action
ssh \$USER@\$HOST	open a secure shell connection as \$USER at \$HOST
ssh -Y \$USER@\$HOST	open a secure shell connection as \$USER at \$HOST allowing to run X11 tunneling
scp \$USER@\$HOST: <i>file dir/</i>	secure copy a file to \$HOST as \$USER
ssh <i>username@dragon2.cs.clemson.edu</i>	Accessing a \$HOST on campus
1. ssh <i>username@access.cs.clemson.edu</i> 2. <i>ssh dragon2.cs.clemson.edu</i>	Accessing a \$HOST from off campus. 1 st ssh into access.cs.clemson.edu, with username. Once logged into access.cs.clemson.edu, you can then ssh into any lab machines (username not necessary)

Methods for Running Linux on Windows

1- Install Ubuntu on Windows 11 using WSL

<https://www.ssl.com/how-to/enable-linux-subsystem-install-ubuntu-windows-10/>

https://www.youtube.com/watch?v=9SugKtGGn_c



2- Install Ubuntu on Windows 11 using VirtualBox tool

<https://www.youtube.com/watch?v=nvdnQX9UkMY>



3- Run linux online:

<https://linuxcontainers.org/incus/try-it/>

<https://labex.io/tutorials/linux-online-linux-playground-372915>